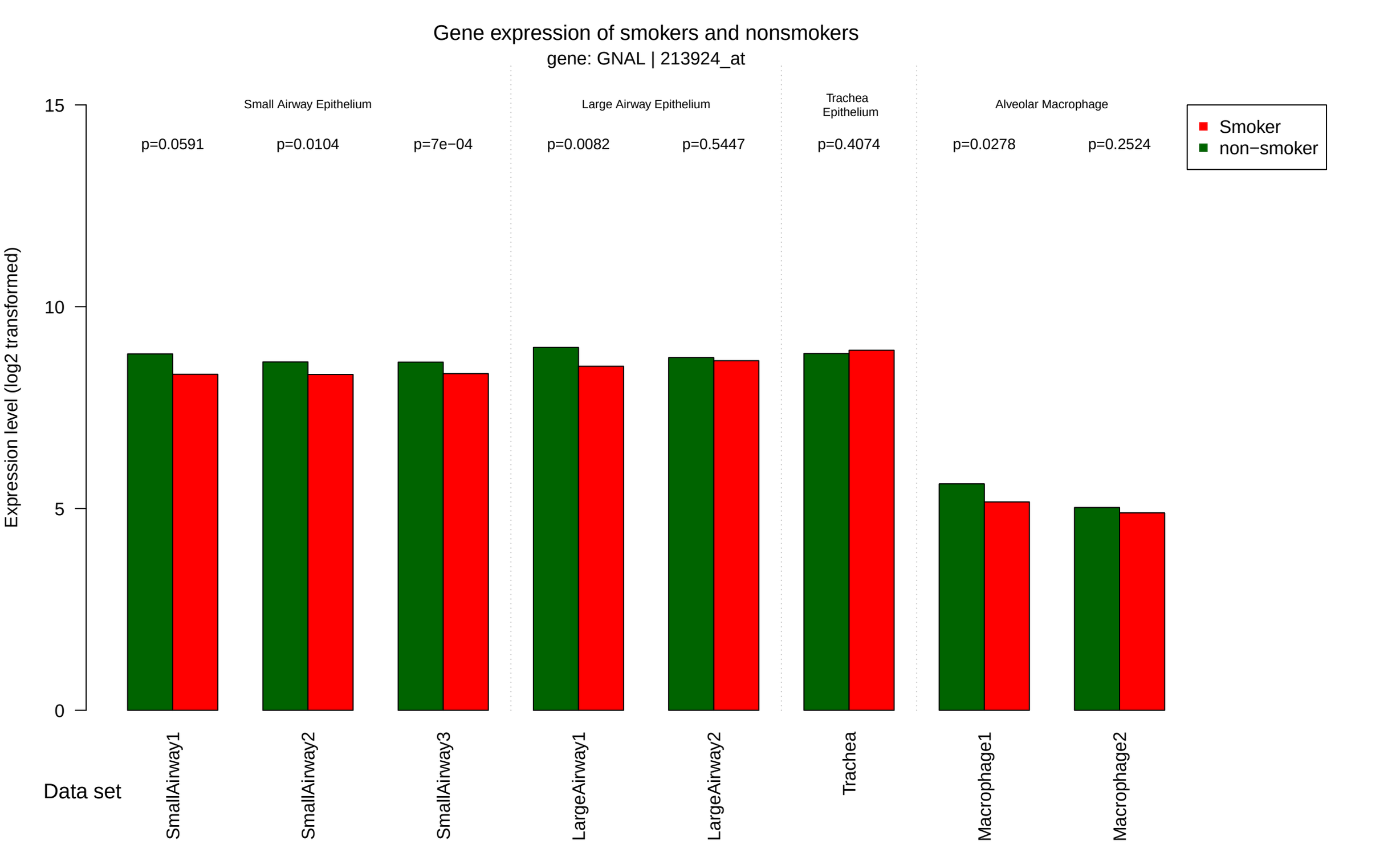
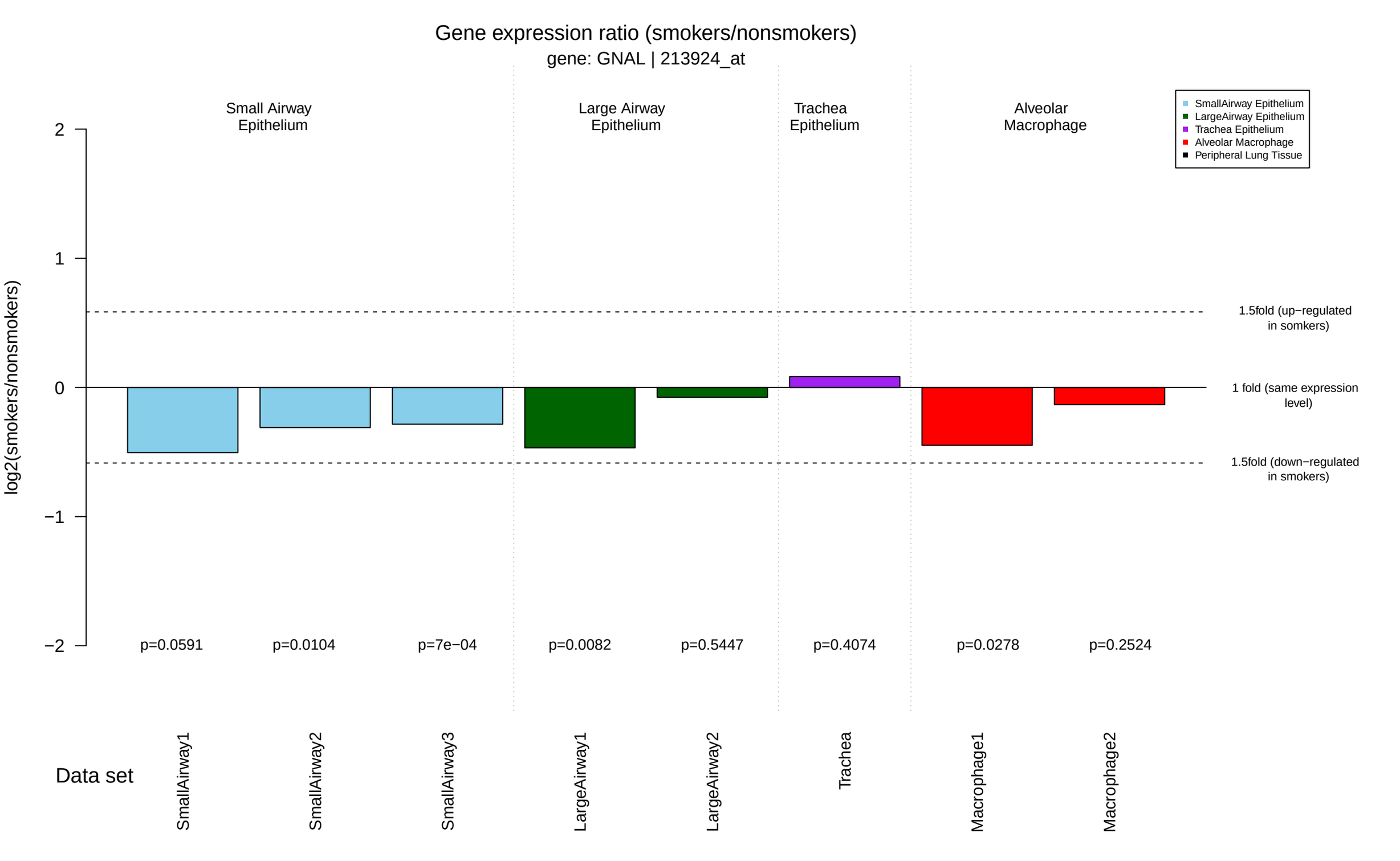
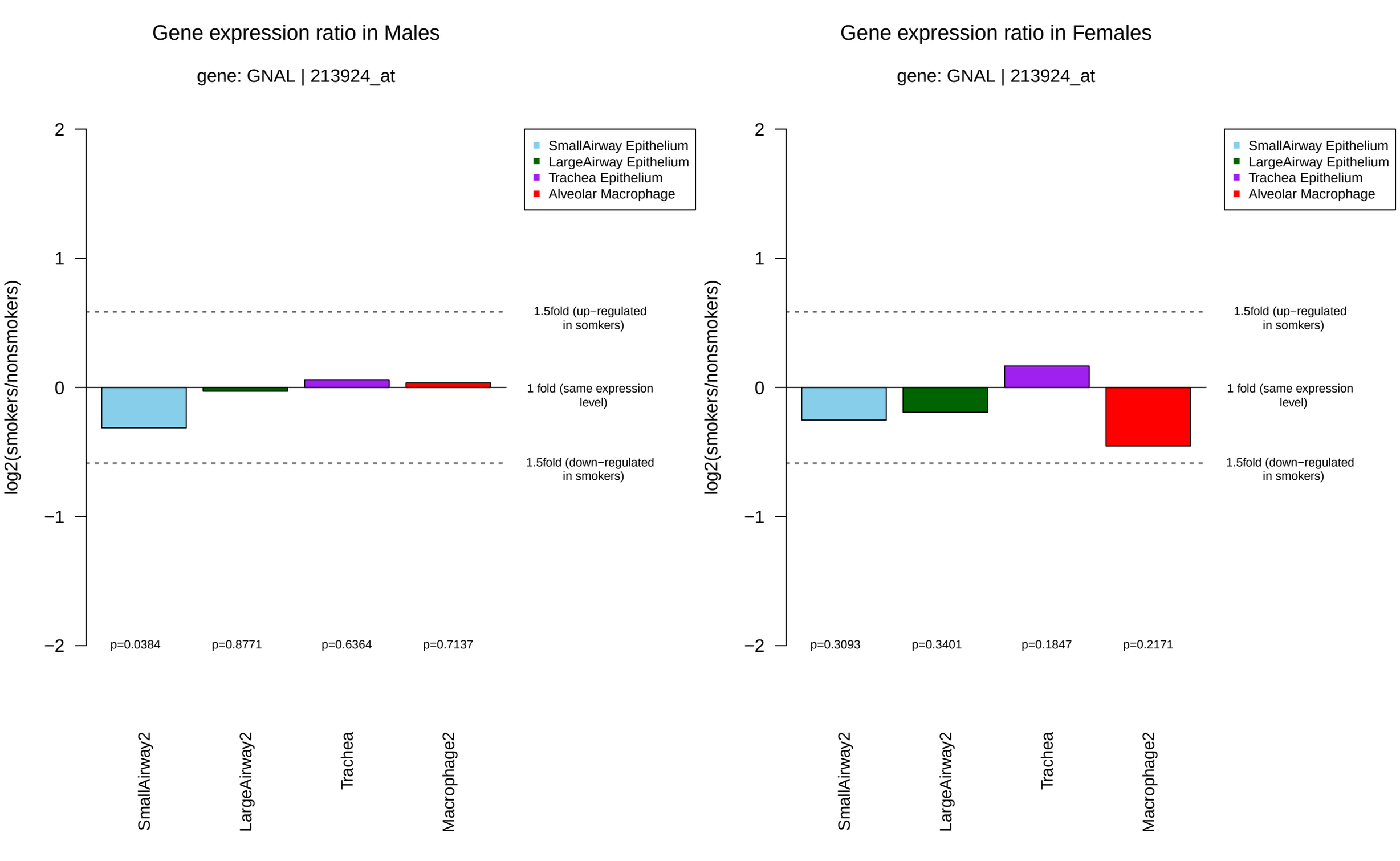
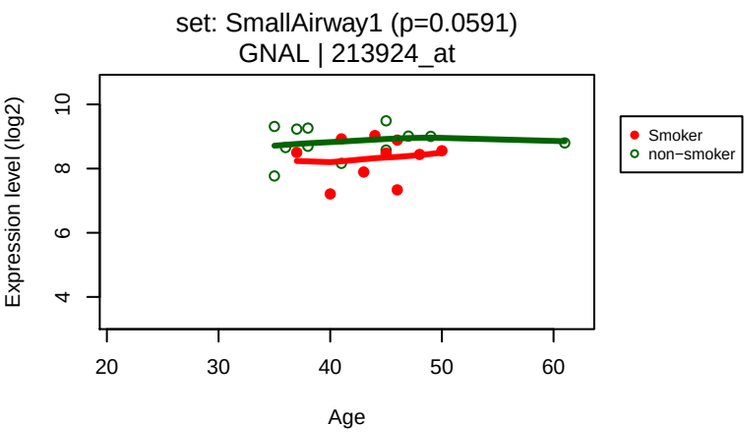
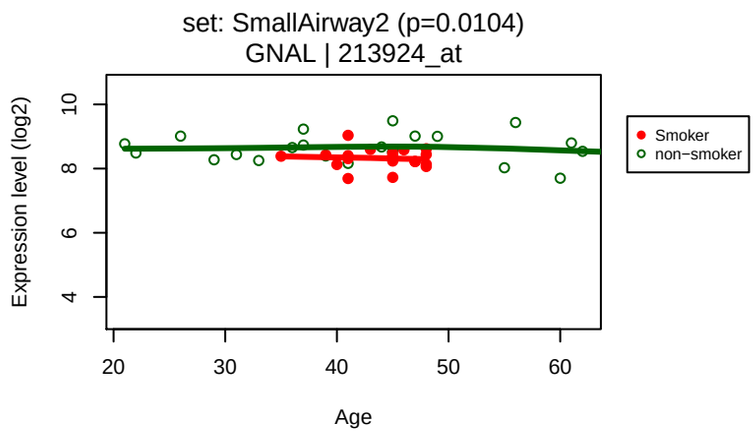
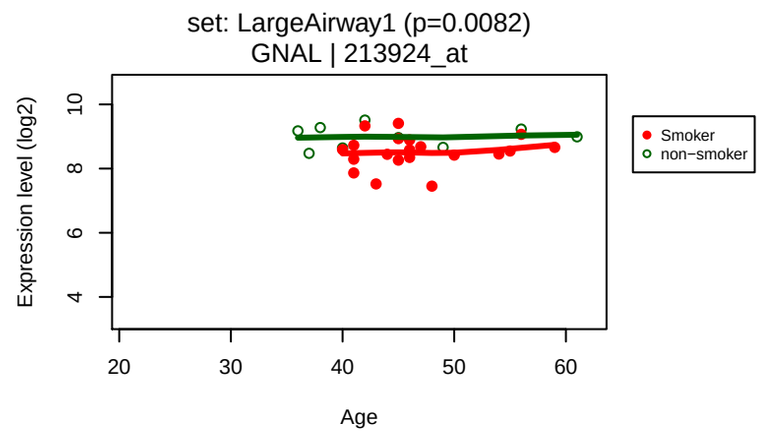

Gene expression of smokers and nonsmokers
gene: GNAL | 213924_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0591
p=0.0104
p=7e−04
p=0.0082
p=0.5447
p=0.4074
p=0.0278
p=0.2524
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

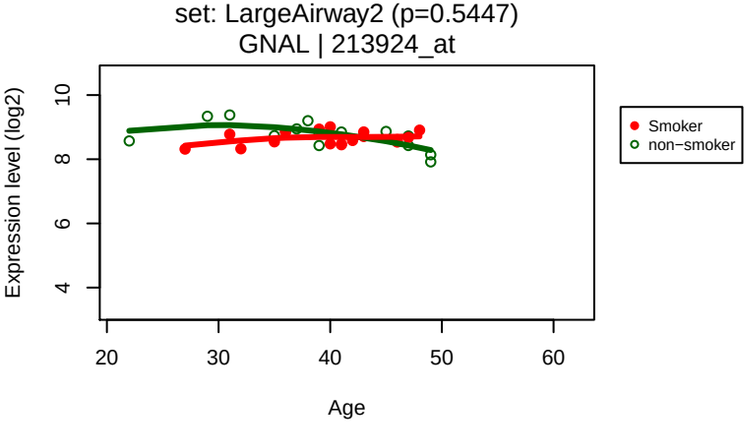

Gene expression ratio (smokers/nonsmokers)
gene: GNAL | 213924_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0591
p=0.0104
p=7e−04
p=0.0082
p=0.5447
p=0.4074
p=0.0278
p=0.2524
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

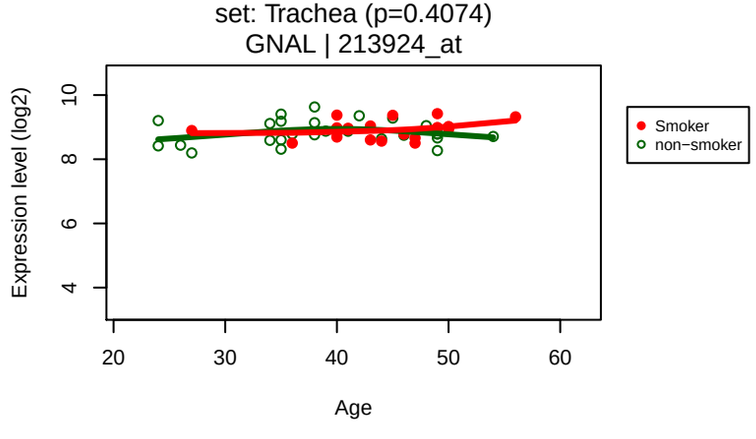

Gene expression ratio in Males
gene: GNAL | 213924_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0384
p=0.8771
p=0.6364
p=0.7137
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAL | 213924_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3093
p=0.3401
p=0.1847
p=0.2171
SmallAirway2
LargeAirway2
Trachea
Macrophage2

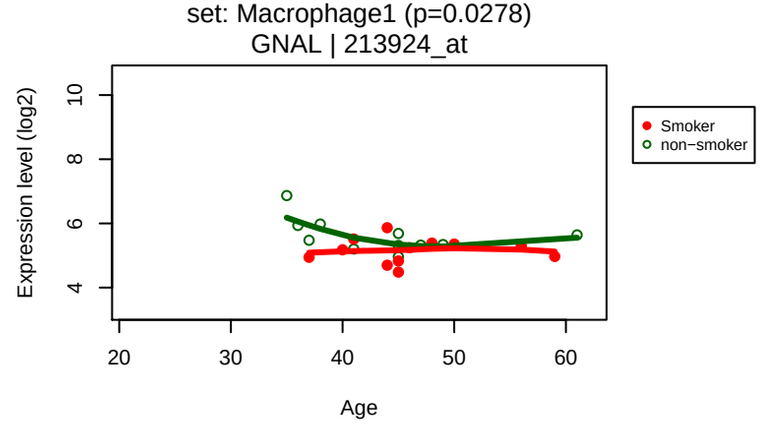

set: SmallAirway1 (p=0.0591)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker

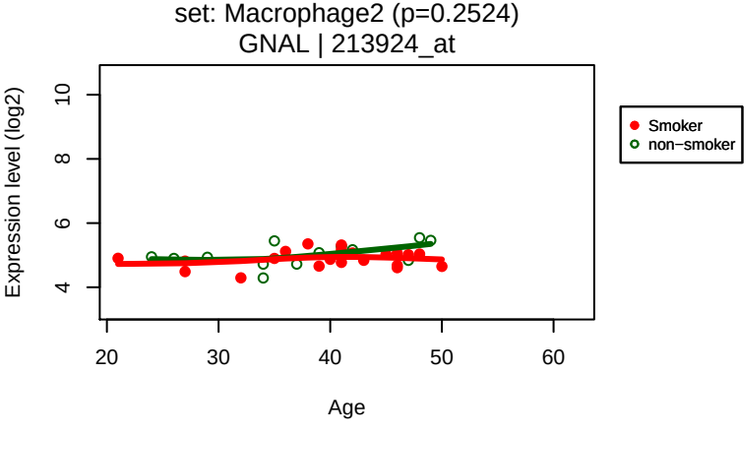

set: SmallAirway2 (p=0.0104)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker

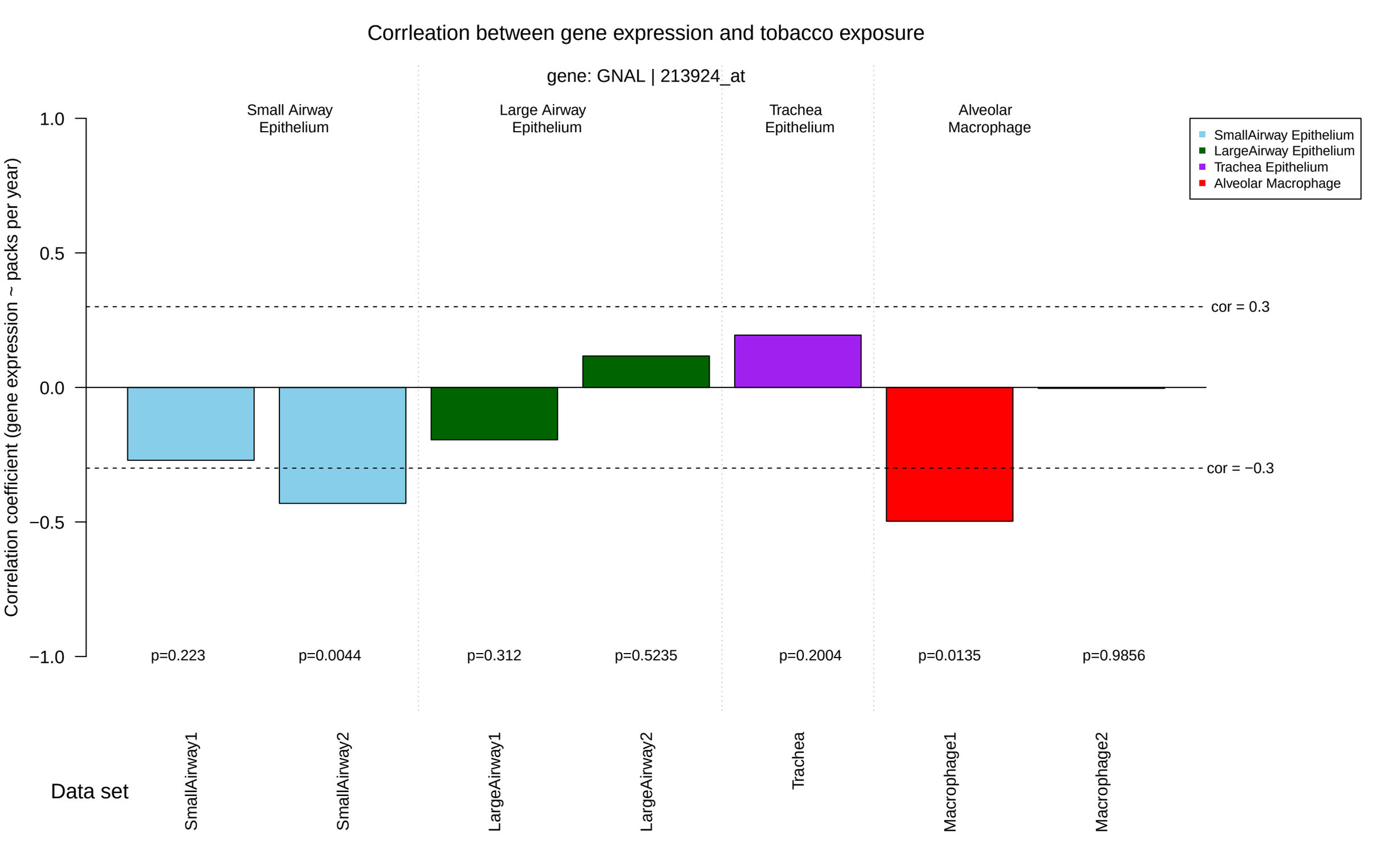

set: LargeAirway1 (p=0.0082)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5447)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4074)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0278)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2524)
GNAL | 213924_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAL | 213924_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.223
p=0.0044
p=0.312
p=0.5235
p=0.2004
p=0.0135
p=0.9856
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2